

JEE MAIN REVISION MASTER: SIMPLE HARMONIC MOTION SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 171 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kinematics & Dynamics of SHM	128 PYQs	SHM Basics
Very High	Energy in SHM & Spring-Pendulum Systems	76 PYQs	Energy & Systems

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Kinematics & Dynamics of SHM	$x(t) = A \sin(\omega t + \phi)$, $v = \omega \sqrt{A^2 - x^2}$, $a = -\omega^2 x$, $T = 2\pi \sqrt{\frac{m}{k}}$
Energy in SHM & Spring-Pendulum Systems	$E_{\text{total}} = \frac{1}{2} m \omega^2 A^2 = \frac{1}{2} k A^2 = \text{const}$, $T_{\text{spring}} = 2\pi \sqrt{\frac{m}{k}}$, $T_{\text{pendulum}} = 2\pi \sqrt{\frac{l}{g}}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kinematics & Dynamics of SHM (Very High Yield — 128 PYQs)

Core Principle: Differential equation of SHM ($d^2x / dt^2 + \omega^2 x = 0$), displacement, velocity, acceleration, and phase relations.

■ **Caution / JEE Pitfall:** Velocity leads displacement by $\pi/2$; acceleration leads displacement by π rad (opposite phase).

■ **Speed Trick:** At mean position ($x = 0$): $v_{\text{max}} = A\omega$, $a = 0$. At extreme ($x = \pm A$): $v = 0$, $a_{\text{max}} = \omega^2 A$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "The frequency of oscillation of a mass m suspended by a spring is ν_1 . If the length of the spring is cut to half, the same mass osci..." [Ans: C]
- [2026 Evening Shift] Q#1 (MCQ): "A uniform disc of radius R and mass M is free to oscillate about the axis A as shown in the figure. For small oscillations the time pe..." [Ans: A]
- [2026 Evening Shift] Q#2 (MCQ): "A spring stretches by 2 mm when it is loaded with a mass of 200 g. From equilibrium position the mass is further pulled down by 2 mm and re..." [Ans: A]

3.2 Energy in SHM & Spring-Pendulum Systems (Very High Yield — 76 PYQs)

Core Principle: Energy oscillation in SHM (frequency of K and U is $2f$), springs in series/parallel, and simple pendulum period.

■ **Caution / JEE Pitfall:** Kinetic energy and potential energy oscillate at TWICE the frequency (2ω) of displacement.

■ **Speed Trick:** When a spring of constant k is cut into ratio $1 : n$, the parts have constants $k_1 = k(n+1)$ and $k_2 = k(n+1 / n)$.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "The frequency of oscillation of a mass m suspended by a spring is $\nu_{(1)}$. If the length of the spring is cut to half, the same mass osci..." **[Ans: C]**
- **[2026 Morning Shift] Q#1 (MCQ):** "A spring of force constant 15N/m is cut into two pieces. If the ratio of their length is $1:3$, then the force constant of smaller piece ..." **[Ans: C]**
- **[2026 Morning Shift] Q#2 (MCQ):** "A simple pendulum of string length 30 cm performs 20 oscillations in 10 s . The length of the string required for the pendulum to perform $40\ldots$ " **[Ans: B]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

The frequency of oscillation of a mass m suspended by a spring is $\nu_{(1)}$. If the length of the spring is cut to half, the same mass oscillates with frequency $\nu_{(2)}$. The value of $\nu_{(2)}/\nu_{(1)}$ is ____.

- (A) 1
- (B) 2
- (C) $\sqrt{2}$ **[Correct]**
- (D) $\sqrt{3}$

Correct Answer: C

Solution Analysis: When a mass m is attached to a spring with spring constant k , the restoring force is given by Hooke's law $F = -kx$. According to Newton's Second Law, $m\frac{d^2x}{dt^2} = -kx \Rightarrow \frac{d^2x}{dt^2} + (k/m)x = 0$. Comparing this to the standard SHM equation $\frac{d^2x}{dt^2} + \omega^2x = 0$, we find the angular frequency as, $\omega = \sqrt{(k/m)}$. Since frequency $\nu = (\omega / 2\pi)$, the frequency of oscillation is: $\nu = (1 / 2\pi)\sqrt{(k/m)} \Rightarrow \nu \propto \sqrt{k}$. A spring constant k is inversely proportional to the natural length L of the spring.

Benchmark Problem 2: JEE Main 2026 (Online) 24th January Morning Shift (Q#1)

A spring of force constant 15N/m is cut into two pieces. If the ratio of their length is $1:3$, then the force constant of smaller piece is ____ N/m .

- (A) 20
- (B) 45
- (C) 60 **[Correct]**
- (D) 15

Correct Answer: C

Solution Analysis: The spring constant k of a spring is inversely proportional to the length L of the spring. $k \propto (1 / L)$. This means that the product of the spring constant and the length is a constant for a given spring: $k \cdot L = \text{Constant}$. The force constant of original spring is $k_{(o)} = 15\text{N/m}$. Let the original length be L . The spring is cut into two pieces with a length ratio of $1:3$. $x + 3x = L \Rightarrow x = (L / 4)$. Hence, the length of smaller spring is $L_{(1)} = x = (L / 4)$ and the length of larger spring is $L_{(2)} = 3x = (3L / 4)$. Let $k_{(1)}$ be the force constant of the smaller piece.